

Customer Behaviour Analysis

Data Analyst Project

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Executive Summary

Customer Behaviour Analysis is an end-to-end data analytics project built to analyze customer purchasing behaviour, identify business trends, and present insights through an interactive Power BI dashboard.

The project used a Kaggle dataset containing approximately 5,050 records and 17 columns, including customer demographics, product details, purchase amount, location, season, payment method, discount status, subscription status, previous purchases, and review ratings. The data was cleaned and prepared using Python Pandas, where missing values, duplicate records, column inconsistencies, and formatting issues were handled.

After cleaning, the dataset was analyzed using PostgreSQL to generate business insights. SQL queries were used to identify high-revenue categories, discount impact, gender-wise revenue, customer segments, subscription behaviour, product ratings, shipping performance, and age-group revenue contribution. The analysis included SQL concepts such as GROUP BY, CASE statements, aggregate functions, and window functions.

The final output was an interactive Power BI dashboard with KPI cards, bar charts, pie charts, and slicers. Key KPIs included Total Revenue, Average Order Value, Total Customers, Subscription Percentage, and Unique Items. The dashboard helped summarize business performance and customer behaviour in a clear visual format.

Major findings showed that the Electronics category generated the highest revenue, customers aged 51+ contributed the highest spending, Debit Card was the most preferred payment method, and New York produced the highest regional revenue. The analysis also showed that high-revenue products did not always receive the highest ratings, which highlights a gap between sales performance and customer satisfaction. Standard and Express shipping were the most preferred delivery options.

Overall, this project demonstrates practical skills in Python, SQL, PostgreSQL, Power BI, data cleaning, business analysis, and dashboard development. It also shows the ability to complete a full analytics workflow from raw data to final business insights.

Project Objective

- To analyze customer purchasing behaviour and identify meaningful business trends using real-world retail data.
- To perform data cleaning and preprocessing using Python Pandas, including handling missing values, removing duplicates, and standardizing data for accurate analysis.
- To understand customer spending patterns, product performance, subscription behaviour, and regional sales trends through exploratory data analysis.
- To practice SQL-based business analysis using PostgreSQL by writing structured queries for revenue analysis, customer segmentation, discount analysis, and product performance evaluation.
- To build an interactive Power BI dashboard for visualizing key business metrics such as Total Revenue, Average Order Value (AOV), Total Customers, and Subscription Percentage.
- To support business decision-making by presenting insights related to customer preferences, payment methods, shipping behaviour, and high-performing products.
- To develop an end-to-end analytics workflow integrating Python, PostgreSQL, and Power BI for practical business analysis.
- To strengthen practical skills in data analytics, dashboard development, business intelligence, and multi-tool integration through a portfolio-based real-world project.

Business Problem

How can a retail company understand customer purchasing behaviour to increase revenue, improve customer retention, and optimize marketing strategies?

Breaking Down the Problem

Low Customer Retention

- Customers are not purchasing frequently

Need to understand:

- Who are frequent buyers?
- Who are one-time buyers?

Ineffective Marketing Campaigns

- Discounts and promo codes may not be effective

Questions:

- Do discounts increase purchases?
- Which customers respond to promo codes?

Product Demand Understanding

- Which category performs best?
- Which products generate the highest revenue?

Payment Behaviour Insights

- Which payment methods are preferred?
- Are digital payments increasing?

Seasonal Demand Variation

- Does season impact sales?
- Which season drives maximum purchases?

Customer Segmentation Problem

Need to identify:

- High-value customers
- Frequent buyers
- Discount-sensitive customers

Dataset Description

Type: Retail / E-commerce Dataset

Nature: Structured (Tabular)

Level: Customer Transaction Level

Data Mix:

- Numerical → Age, Purchase Amount, Previous Purchases
- Categorical → Category, Payment Method, Season

This dataset is used for customer behaviour analysis + business decision-making

Column Name	Description
Customer ID	Unique identifier for each customer
Age	Age of the customer
Gender	Gender of the customer (Male/Female/Other)
Item Purchased	Specific product purchased
Category	Product category (Clothing, Electronics, etc.)
Purchase Amount	Amount spent by the customer
Location	Customer’s geographic location
Size	Product size (S, M, L, etc.)
Color	Product color
Season	Season of purchase (Winter, Summer, etc.)
Review Rating	Customer rating for the product
Subscription Status	Whether customer is subscribed (Yes/No), Subscription Program
Shipping Type	Type of shipping selected

Discount Applied	Whether discount was applied
Previous Purchases	Number of past purchases
Payment Method	Mode of payment
Frequency of Purchases	Purchase frequency (Weekly, Monthly, etc.)

EDA- (Exploratory Data Analysis using Python)

Data Loading

Imported pandas

Loaded dataset using:

```
#importing the dataframe/dataset  
df=pd.read_csv('customer_shopping_behavior.csv')
```

Why was this step performed?

To bring raw data into a structured DataFrame

Enables further analysis and transformations

df.head()

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
0	2701	22	Female	T-shirt	Clothing	68.0	California	XL	Olive	Winter	3.2	No	Standard	No	No	36.0	Cash
1	521	51	Male	Sunglasses	Accessories	84.0	South Carolina	M	White	Spring	3.9	Yes	Free Shipping	Yes	Yes	20.0	Debit Card
2	3157	18	Female	Shirt	Clothing	50.0	Montana	M	Black	Winter	3.1	No	2-Day Shipping	No	No	18.0	Cash
3	1687	22	Male	Gloves	Accessories	75.0	Illinois	L	Red	Fall	4.2	No	Store Pickup	No	No	25.0	Cash
4	2929	40	Female	Jewelry	Accessories	80.0	Alabama	L	Yellow	Spring	3.6	No	Store Pickup	No	No	17.0	Credit Card

Initial Data Exploration

Activities Performed:

Viewed data using:

- df.head()
- df.tail()

Checked dataset summary:

- df.describe(include='all')
- df.info()

Checked missing values:

- `df.isnull().sum()`

Why was this step performed?

To understand:

- Data structure
- Data types
- Missing values

This step helps identify data quality issues early

Data Consistency Check (Category Correction)

Activities Performed:

Checked unique values in:

- Item Purchased
- Category
- Found inconsistencies between Item Purchased and Category
- Created a mapping dictionary
- Updated category using:

Why was this step performed?

- Categories were inconsistent or incorrect
- Ensures:
 - Accurate grouping
 - Correct business insights

This is a critical real-world data cleaning step

Handling Missing Values (Size Column)

Activities Performed:

Handled missing values based on category:

- Electronics → "Not Applicable."
- Accessories → "Not Applicable."
- Footwear → "Free Size."
- Clothing → Filled with mode

Why was this step performed?

- Different categories require different business logic
- Example:

- Electronics don't have size → Not Applicable
- Clothing → size is important → use mode

This shows domain understanding

Handling Missing Values (Review Rating)

Activities Performed:

- Checked:
 - Mean, median, mode
- Filled missing values using:
- Rounded values

Activities Performed:

- Instead of the global average, the product-level mean is used
- This keeps ratings realistic and meaningful

This is an advanced imputation technique

Handling Missing Values (Purchase Amount)

Activities Performed:

Replaced with the mean value. Why did you do it?

- Maintains product-level pricing consistency
- Avoids distortion in revenue analysis

Handling Missing Values (Previous Purchases)

Activities Performed:

Replaced with the 0 (zero) value.

Activities Performed:

- Missing values imply no previous purchase
- Important for:
 - Customer segmentation
 - Loyalty analysis

Duplicate Handling

Activities Performed:

- Checked duplicates:
- Identified duplicates using Customer ID
- Removed duplicates:

Why was this step performed?

Ensures:

- Each customer appears only once
- No double-counting in analysis

Critical for accurate reporting

Column Standardization

Activities Performed:

- Converted column names:

```
df.columns = df.columns.str.replace(" ", "_") df.columns = df.columns.str.lower()
```

- Renamed column:

Renaming the Headers

```
df.columns=df.columns.str.replace(" ", "_")  
df.columns=df.columns.str.lower()
```

Why did you do it?

- Makes column names: Clean
 - SQL-friendly
 - Easy to use in Power BI

Data Analysis using SQL Server

1. Which category generates highest revenue?

	category	total_revenue
1	Electronics	486366.96
2	Accessories	194866.87
3	Clothing	183188.39
4	Footwear	111236.2
5	Outerwear	18524

2. Are discounts actually increasing purchase value?

discount_applied	total_revenue	Avg_Purchase
No	502241.64	182.24
Yes	491940.78	219.22

3. What is the total revenue generated by male and female customer?

gender	total_revenue
Male	390964.526238359
Female	352060.293140485
Other	251157.60467728

4. Which customer used a discount but still spent more than the average purchase amount?

	customer_id	purchase_amount	discount_applied
1	4568	276.343789954338	Yes
2	4256	734.47734939759	Yes
3	4289	1282.44	Yes
4	4576	831.344565217391	Yes
5	4319	685.231470588235	Yes
6	4708	735.773625	Yes
7	4584	685.231470588235	Yes
8	4607	557.24	Yes
9	4851	685.231470588235	Yes
10	4845	692.11	Yes

5. Which is the top/bottom 5 products with the highest average review rating.

	item_purchased	avg_review_rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.8
5	Skirt	3.79

	item_purchased	avg_review_rating
1	Phone	2.94
2	Headphones	3.08
3	Watch	3.1
4	Bag	3.17
5	Laptop	3.19

6. Average purchase: Standard vs Express shipping

shipping_type	total_revenue	total_orders
Express	358.12	1199
Standard	340.57	1201

7. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

subscription_status	total_customers	total_revenue	avg_spend
Yes	1579	427245.4	270.58
No	3421	566937.02	165.72

8. Top 5 products with highest discount usage %

item_purchased	total_number_of_times_sold	Number_of_times_sold_when_discount_applied	discount_percent
Laptop	143	74	51.748251748251
Phone	171	88	51.461988304093
Headphones	166	85	51.204819277108
Hat	154	77	50.000000000000
Watch	157	78	49.681528662420

9. Segment customer into new, returning and loyal based on their total number of previous purchases, and show the count of each segment.

customer_segment	Customer_Count
Loyal Customers	3085
New Customer	561
Returning Customer	1354

10. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

category	item_purchased	most_purchased	RNK
Accessories	Jewelry	171	1
Accessories	Bag	164	2
Accessories	Belt	161	3
Accessories	Sunglasses	161	3
Clothing	Shirt	315	1
Clothing	Blouse	171	2
Clothing	Pants	171	2
Electronics	Phone	171	1
Electronics	Headphones	166	2
Electronics	Watch	157	3
Footwear	Shoes	303	1
Footwear	Sandals	160	2
Footwear	Sneakers	145	3
Outerwear	Jacket	163	1
Outerwear	Coat	161	2

11. What are the top 3 most purchased products within each category?

Customer_type	subscription_status	Customer_count
Repeat Buyers	No	2763
Normal Buyers	Yes	389
Normal Buyers	No	658
Repeat Buyers	Yes	1190

12. What is the revenue contribution of each age group?

age_group	total_revenue
51+	399345.32
35-50	266706.7
26-35	177421.93
18-25	150708.48

Power BI Dashboard



[Customer Revenue Analysis]



[Best – Worst Product Analysis]

Key Business Insights / Findings

- The Electronics category generated the highest overall revenue, indicating strong customer demand for technology-related products.
- Customers aged 51 and above contributed the highest revenue share, showing higher purchasing power and spending behaviour among older customers.
- Debit Card was the most preferred payment method, followed by Cash and Credit Card transactions, indicating strong usage of digital payment systems.
- Standard and Express shipping methods were the most frequently used delivery options, suggesting customer preference for reliable and faster shipping services.
- Some high-revenue products such as Headphones and Phones generated strong sales despite having comparatively lower customer review ratings. This highlights a gap between sales performance and customer satisfaction.
- Jewellery and Pants received the highest average review ratings, indicating strong customer satisfaction and product quality perception.
- New York generated the highest regional revenue, followed by Los Angeles and Phoenix, making these locations important target markets for business growth and marketing strategies.
- Customers who made repeat purchases showed a higher likelihood of subscription participation, indicating strong customer loyalty and engagement.
- Discount strategies influenced customer purchasing behaviour, as several discounted purchases still exceeded the average purchase amount.
- Customer segmentation analysis identified different customer groups such as new, returning, and loyal customers, helping businesses understand customer retention behaviour and spending patterns.

Future Enhancements

- Integrate real-time data sources to build a live customer analytics dashboard.
- Implement machine learning models for customer segmentation, product recommendation, and purchase prediction.
- Add sales forecasting and trend prediction features using historical purchase data.
- Include advanced Power BI visualizations and drill-through reports for deeper business analysis.
- Expand the dataset with customer feedback, product inventory, and marketing campaign data for broader analysis.
- Automate the ETL (Extract, Transform, Load) process for continuous data updates and reporting.
- Develop customer churn analysis to identify customers likely to stop purchasing.
- Deploy the dashboard on cloud platforms for easier business accessibility and collaboration.

Conclusion

The “Customer Behaviour Analysis” project successfully demonstrated an end-to-end data analytics workflow using Python, PostgreSQL, and Power BI. The project focused on analyzing customer purchasing behavior, identifying sales trends, and generating business insights from retail customer data.

The dataset was cleaned and prepared using Python Pandas by handling missing values, removing duplicates, and performing consistency checks to improve data quality. PostgreSQL was used to perform structured business analysis through SQL queries related to revenue generation, customer segmentation, product performance, discount analysis, and subscription behavior.

An interactive Power BI dashboard was developed to visualize key business metrics such as Total Revenue, Average Order Value, Customer Distribution, Product Performance, and Payment Preferences. The dashboard provided clear and user-friendly visual insights to support business understanding and decision-making.

The analysis revealed important findings such as high-performing product categories, customer age-group spending trends, preferred payment methods, and regional sales distribution. Overall, the project helped strengthen practical skills in data cleaning, SQL querying, dashboard development, and business-focused analytics while demonstrating the complete lifecycle of a real-world data analysis project.

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Thank You!